

CSC1122 Lab 2: Text Classification

1 Naive Bayes from Scratch

1.1 Training

Load the training split from the Nigerian English news classification dataset. Click on the “Use This Dataset” button for code snippets to load the data into the pandas or Datasets format. Write code to calculate the prior probability of the six classes. For each word (type not token) in the training data, calculate the probability of the word given each class. Note that you can use just the title, just the excerpt or the concatenation of the title and the excerpt.

1.2 Testing

- Load the test split of the same dataset.
- Calculate the news category of each instance in the test set.
- Calculate the accuracy of your classifier.

2 Naive Bayes Scikit-learn

Replace the code that you wrote above with Scikit-learn’s Naive Bayes implementation.